

Study of Effect of Heat on Ice

Aim

To study the effect of heat on ice and plot the temperature versus time graph.

Requirements

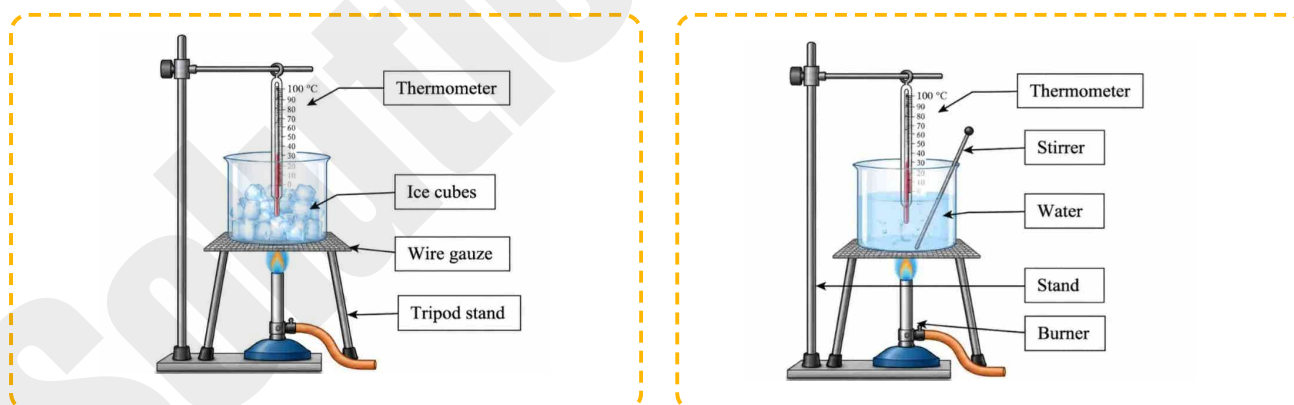
A beaker, ice cubes, a thermometer, a stand, a burner, a tripod, a wire gauze, and a stirrer.

Procedure

1. Determine the least count of the thermometer.
2. Place a few ice cubes into a glass beaker.
3. Set the tripod on the stand, place the wire gauze on the tripod, and keep the beaker on top of it. Insert the bulb of the thermometer into the ice and note down its temperature.
4. Gently heat the beaker using the burner.
5. Note the temperature of the ice/water after every minute.
6. As the ice is heated, it starts to melt — stir the ice-water mixture well, and continue heating even after all the ice has melted.
7. Keep recording the temperature of the water even after it starts boiling.
8. Plot the graph of temperature versus time.

Diagrams

(Label the various components in the given diagram)



Observations

1. Least count of the thermometer = 1°C .
2. The temperature of the mixture of ice and water remains 0°C till the ice melts completely.

- As heating is continued, the temperature goes on increasing from 0°C to 100°C.
- The temperature of water remains constant (100°C) even after it starts boiling.

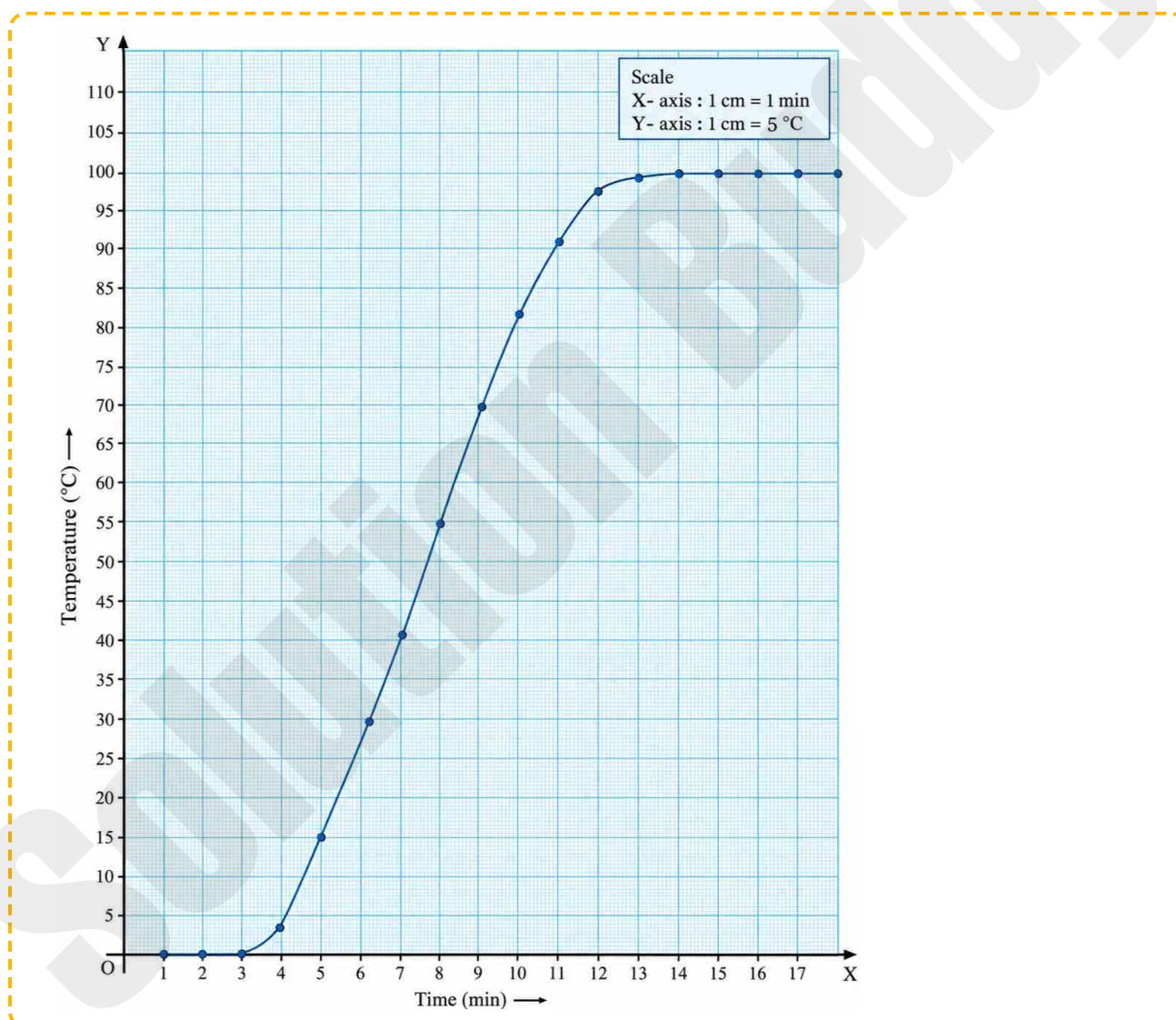
Observation Table

Time (min)	0	1	2	3	4	5	6	7	8	9	10	11	12	13
Temperature (°C)	0	0	0	0	1.5	15	29	43	57	71	83	94.5	98	99

Time (min)	14	15	16	17	18	19	20	21	22	23	24	25	26	27
Temperature (°C)	100	100	100	100										

Time (min)	28	29	30	31	32	33	34	35	36	37	38	39	40
Temperature (°C)													

Graph of Temperature versus Time



Scale — X-axis: 1 cm = 1 min | Y-axis: 1 cm = 5°C

Inference / Conclusion

The thermometer shows a rise in temperature only after all the ice has melted. The thermometer does not show a rise in temperature once the water starts boiling. From these observations, we conclude that heat energy is absorbed during the transformation of ice into water and water into steam, but there is no rise in temperature during the transformation state. This is indicated by the constant reading of the thermometer.

Precaution

The bulb of the thermometer should not touch the bottom of the beaker.

Multiple Choice Questions

1. When ice melts, its volume

- (A) increases (B) decreases (C) remains constant (D) increases or decreases

Answer: (B) decreases

2. Ice/water is a substance that

- (A) expands on melting and contracts on freezing
(B) contracts on melting and does not undergo change in volume on freezing
(C) contracts on melting and expands on freezing
(D) does not undergo any change in volume on melting or freezing

Answer: (C) contracts on melting and expands on freezing

3. Heat absorbed when 1g of ice melts at 0°C to form 1g of water at the same temperature is cal.

- (A) 80 (B) 800 (C) 540 (D) 54

Answer: (A) 80

4. Heat needed to raise the temperature of 1kg of water from 14.5°C to 15.5°C is

- (A) 4180 J (B) 10^3 J (C) 1 cal (D) 4180 cal

Answer: (A) 4180 J

5. Heat needed to convert 1g of water at 100°C and at a pressure of one atmosphere into 1g of steam under the same conditions is

- (A) 80 cal (B) 540 cal (C) 89 J (D) 540 J

Answer: (B) 540 cal

Remember

When there is a change of state of a substance, solid $\leftrightarrow$ liquid, liquid $\leftrightarrow$ gas (vapour), solid $\leftrightarrow$ gas (vapour), heat energy is absorbed by the substance or heat energy is

removed from the substance at constant temperature.

Date: _____

Teacher's Signature: _____